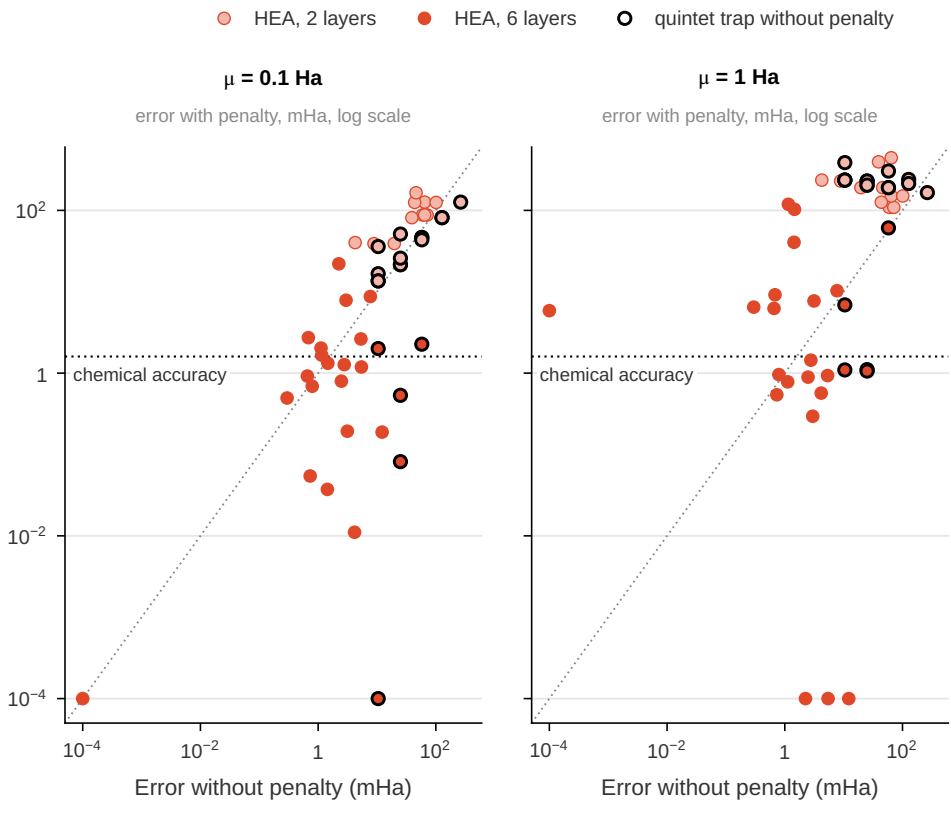
Error with vs without a spin penalty, HEA on the H_4 chain



H_4 chain, STO-3G, noiseless statevector. Cost E + μ S^2; each point is one start, run with and without the penalty from the same parameters (d = 1.5-2.5 Å, 5 starts). Below the dotted diagonal the penalty lowered the error. Errors below 10^{-4} mHa are drawn at the floor.